



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - DEC 2023

UE20CS352 - Object Oriented Analysis & Design with Java

Total Marks : 100.0

1.a. State any four differences between Method overloading and Method overriding? (4.0 Marks)

1.b. State the reason why the given code prints output as false. Also change the code such that the output in the main function is true.

```
public class MainClass
{
    public static void main(String[] args)
    {
        Classdemo objOne=new Classdemo(10,20);
        Classdemo objTwo=new Classdemo(10,20);
        System.out.println(objOne.equals(objTwo)); //prints false
    }
}
class Classdemo
{
    int a; int b;
    Classdemo(int a,int b)
    {
        this.a=a;
        this.b=b;
    }
}
```

(5.0 Marks)

1.c. Elucidate need for an abstract class and justify the same with an example. (6.0 Marks)

1.d. **Give the output of java program given below:**

```
public class StaticDemo {  
    static int a=1,b;  
    static void display(int x){  
        System.out.println("x="+x);  
        System.out.println("a="+a);  
        System.out.println("b="+b);  
    }  
    static{  
        System.out.println("Inside static block");  
        b=a*2;  
    }  
    public static void main(String[] args) {  
        System.out.println("Inside main");  
        display(5);  
    }  
}
```

(5.0 Marks)

2.a. Elucidate qualified association and aggregation relationship w.r.t class diagram with an example for each. (6.0 Marks)

2.b. Draw a Use Case diagram of a Railway Reservation System. In the system, a Passenger logs to the system and checks for trains available. The passenger can book a ticket and optionally choose a seat of their choice and accordingly make a card payment for it. The bank validates the payment for the system. Passengers can also cancel their ticket. The system admin processes the cancellation as well as update train when required. (6.0 Marks)

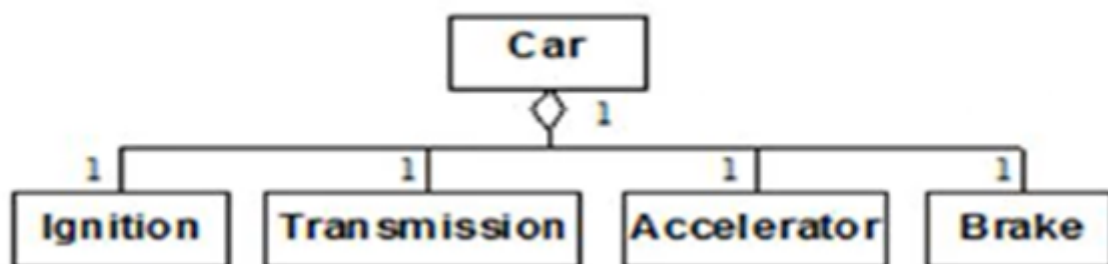
2.c. Describe in a line or two the following elements of activity diagram with their UML notation:

i) Activity

ii) Object Flow

(4.0 Marks)

2.d. Draw a state diagram for Transmission component of a Car wrt the diagram given below



(4.0 Marks)

3.a. Explain the creator and controller GRASP principles with an example along with the corresponding class diagram (8.0 Marks)

3.b. Define MVC architectural pattern. Explain the working of MVC with a neat diagram (6.0 Marks)

3.c. Fill in the blanks:

- i. _____ Violating Dependency Inversion Principle will result in a code with _____ coupling.
- ii. _____ Liskov Substitution Principle states that the designer/implementer should be able to replace objects of _____ with the objects of _____.
- iii. _____ Principle states that the design should be extensible however modifying existing design should be avoided.
- iv. _____ The pattern that hides the complexity related to creation of objects from the client is _____.
- v. A _____ is a best practice, a recipe, a time tested solution to a recurring problem that's proved to work and to be the best one among to choose. (6.0 Marks)

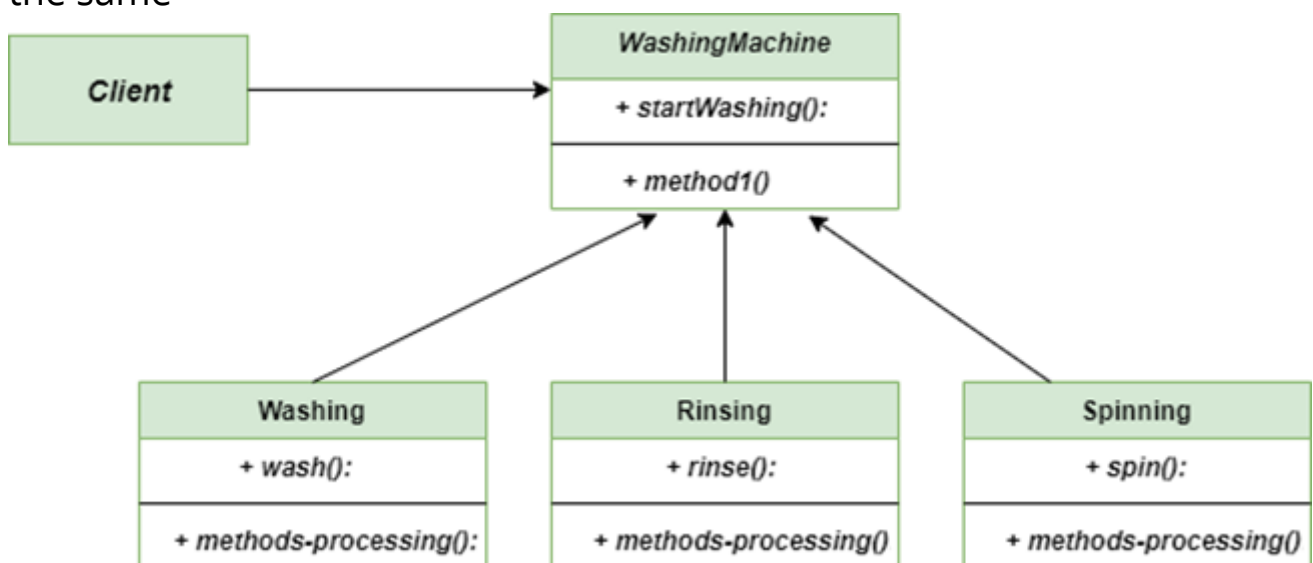
4.a. State the difference between Architectural and Design patterns
(6.0 Marks)

4.b. Implement the following use case using a proxy pattern. Write both the class diagram and the code snippet. State which type of proxy pattern is suitable for the given use case.

Use Case: If an entire library of books, with all the details about each book, is loaded from a database, it will consume a lot of RAM, and it is very likely that the user will need to issue or return only one book. The solution, that uses the proxy design pattern, displays only the name, author, and availability of the books when a list of them is displayed. When a book is selected, all of the remaining details will be fetched from the database and operations like issue/return (update availability) can be performed.

Note: write only code snippets/pseudo code
(6.0 Marks)

4.c. Which design pattern is depicted in the given diagram below and describe the pattern in a line or two. Give the pseudo code to implement the same



(8.0 Marks)

5.a. When to use Behavioral patterns? Describe in a line or two interpreter and iterator pattern (4.0 Marks)

5.b. What is an Anti pattern? Briefly describe vendor lock-in anti pattern. (6.0 Marks)

5.c. For the given scenario identify which behavioral pattern can be used? Briefly describe the chosen pattern and draw the class diagram for the scenario with the chosen pattern. (2+3+5)

Scenario

An enterprise has been getting more email than they can handle. The enterprise gets 4 types of e-mail.

They are:

- Fan mail with compliments,
- Complaints,
- Requests for new features,
- Spam.

Your task:

- The enterprise has already written the AI detectors that can tell whether an email is fan, complaint, request or spam,
- You need to create a design that can use the detectors to handle incoming email.

(10.0 Marks)